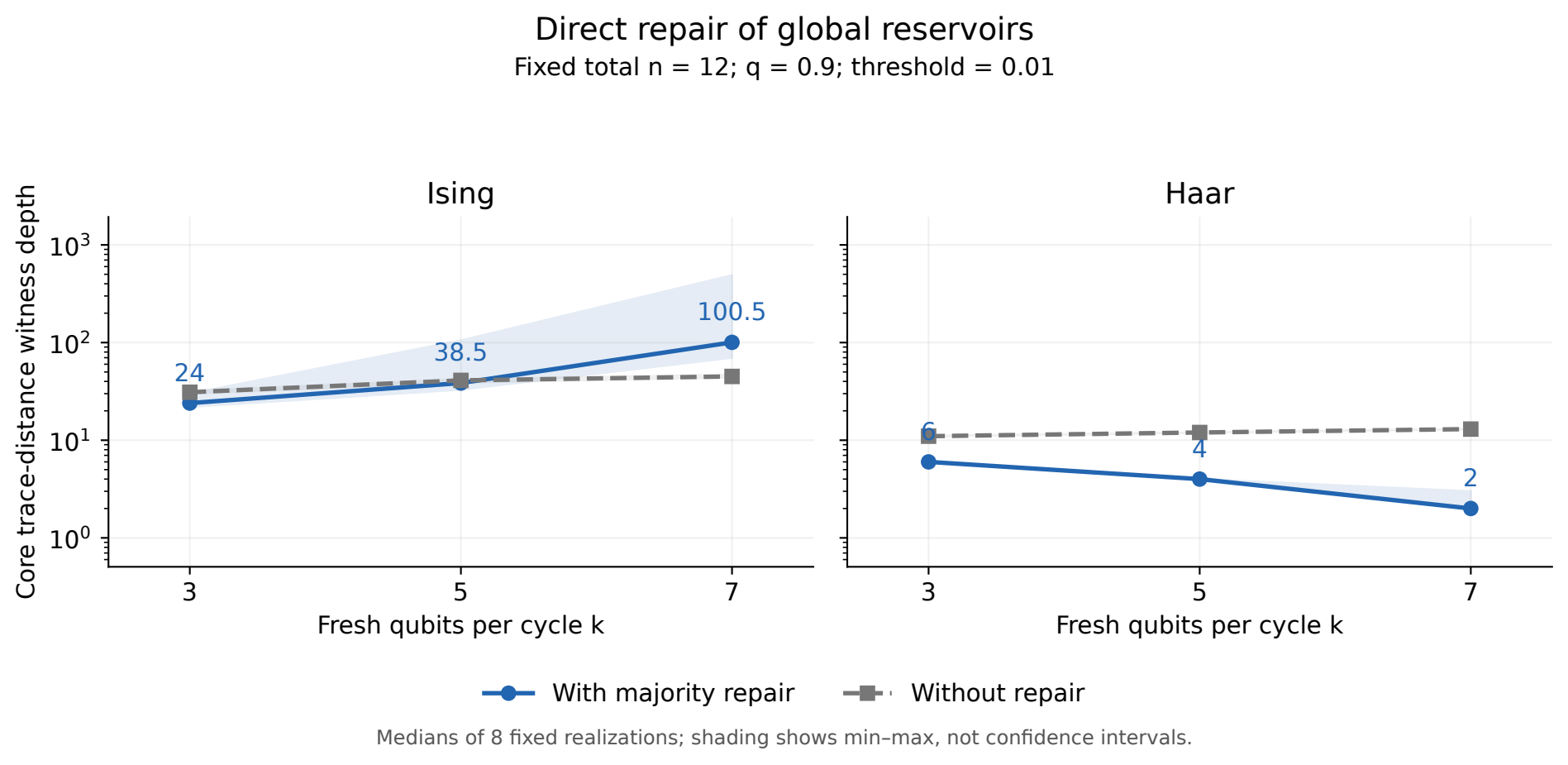

Direct repair of global reservoirs
Fixed total n = 12; q = 0.9; threshold = 0.01
Ising
Haar
Core trace-distance witness depth
$10^3$
$10^2$
$10^1$
$10^0$
24
38.5
100.5
6
4
2
3
5
7
Fresh qubits per cycle k
With majority repair
Without repair
Medians of 8 fixed realizations; shading shows min–max, not confidence intervals.